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MCY 612
Lab 5 Logging
April, 2025

**MCY 612 Cloud Security**
**Lab#5: Logging**

In this lab, you will evaluate an ELB system via its access log file.

**Tools and Materials**

AWS credentials
AWS CloudShell CLI
Ubuntu Server 16.04 LTS AMI: ami-0ee02acd56a52998e

**What is the id of the VPC that you created in lab 2?**
**What is the id of the subnet that you created in lab 2?**
**What is the id of the security group that you created in lab 2?**

Note: You need to replace words in italics with your information.

**Part 1: Configure ELB Log**

Start your AWS CloudShell terminal.

1.Modify the security group you created in lab2 to allow 80 traffic:

aws ec2 authorize-security-group-ingress --group-id *your_security_group_id* --protocol tcp --port 80 --cidr 0.0.0.0/0

**What is your command and its complete output?**

```
~ $ aws ec2 authorize-security-group-ingress --group-id sg-0b1cfa43fd7ea602b --protocol tcp --port 80 --cidr 0.0.0.0/0
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-0c1dfa8ca1e452535",
            "GroupId": "sg-0b1cfa43fd7ea602b",
            "GroupOwnerId": "881490118823",
            "IsEgress": false,
            "IpProtocol": "tcp",
            "FromPort": 80,
            "ToPort": 80,
            "CidrIpv4": "0.0.0.0/0",
            "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:881490118823:security-group-rule/sgr-0c1dfa8ca1e452535"
        }
    ]
}
:...skipping...
:...skipping...
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-0c1dfa8ca1e452535",
            "GroupId": "sg-0b1cfa43fd7ea602b",
            "GroupOwnerId": "881490118823",
            "IsEgress": false,
            "IpProtocol": "tcp",
            "FromPort": 80,
```

```
            "ToPort": 80,
            "CidrIpv4": "0.0.0.0/0",
            "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:881490118823:security-group-rule/sgr-0c1dfa8ca1e452535"
        }
    ]
}
~
(END)...skipping...
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-0c1dfa8ca1e452535",
            "GroupId": "sg-0b1cfa43fd7ea602b",
            "GroupOwnerId": "881490118823",
            "IsEgress": false,
            "IpProtocol": "tcp",
            "FromPort": 80,
            "ToPort": 80,
            "CidrIpv4": "0.0.0.0/0",
            "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:881490118823:security-group-rule/sgr-0c1dfa8ca1e452535"
        }
    ]
}
~
~
~
~
(END)...skipping...
```

```
(END)...skipping...
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-0c1dfa8ca1e452535",
            "GroupId": "sg-0b1cfa43fd7ea602b",
            "GroupOwnerId": "881490118823",
            "IsEgress": false,
            "IpProtocol": "tcp",
            "FromPort": 80,
            "ToPort": 80,
            "CidrIpv4": "0.0.0.0/0",
            "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:881490118823:security-group-rule/sgr-0c1dfa8ca1e452535"
        }
    ]
}
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~
(END)...skipping...
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-0c1dfa8ca1e452535",
```

```
            "GroupId": "sg-0b1cfa43fd7ea602b",
            "GroupOwnerId": "881490118823",
            "IsEgress": false,
            "IpProtocol": "tcp",
            "FromPort": 80,
            "ToPort": 80,
            "CidrIpv4": "0.0.0.0/0",
            "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:881490118823:security-group-rule/sgr-0c1dfa8ca1e452535"
        }
    ]
}
~
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~
(END)...skipping...
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-0c1dfa8ca1e452535",
            "GroupId": "sg-0b1cfa43fd7ea602b",
            "GroupOwnerId": "881490118823",
            "IsEgress": false,
            "IpProtocol": "tcp",
```


AWS


```
            "FromPort": 80,
            "ToPort": 80,
            "CidrIpv4": "0.0.0.0/0",
            "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:881490118823:security-group-rule/sgr-0c1dfa8ca1e452535"
        }
    ]
}
~
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~
~
~
~
~
(END)...skipping...
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-0c1dfa8ca1e452535",
            "GroupId": "sg-0b1cfa43fd7ea602b",
            "GroupOwnerId": "881490118823",
            "IsEgress": false,
            "IpProtocol": "tcp",
            "FromPort": 80,
            "ToPort": 80,
            "CidrIpv4": "0.0.0.0/0",
```


AWS CloudShell

```
            "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:881490118823:security-group-rule/sgr-0c1dfa8ca1e452535"
        }
    ]
}
~
~
~
~
~
~
~
~
~
~
(END)...skipping...
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-0c1dfa8ca1e452535",
            "GroupId": "sg-0b1cfa43fd7ea602b",
            "GroupOwnerId": "881490118823",
            "IsEgress": false,
            "IpProtocol": "tcp",
            "FromPort": 80,
            "ToPort": 80,
            "CidrIpv4": "0.0.0.0/0",
            "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:881490118823:security-group-rule/sgr-0c1dfa8ca1e452535"
        }
```

2.Create a load balancer via the following command:

aws elb create-load-balancer --load-balancer-name your_username --listeners "Protocol=HTTP,LoadBalancerPort=80,InstanceProtocol=HTTP,InstancePort=80" --subnets your_subnet_id --security-groups your_security_group_id

**What is your command and its complete output?**

```
~ $ aws elb create-load-balancer --load-balancer-name carrd12 --listeners "Protocol=HTTP,LoadBalancerPort=80,InstanceProtocol=HTTP,InstancePort=80" --subnets su
bnet-09df9a6645aea6ede --security-groups sg-0b1cfa43fd7ea602b
{
    "DNSName": "carrd12-1091338731.us-east-1.elb.amazonaws.com"
}
~ $
```

3.Describe your load balancer with the following command:

aws elb describe-load-balancers --load-balancer-name your_username

**What is the load balancer DNS Name?**

"DNSName": "carrd12-1091338731.us-east-1.elb.amazonaws.com",

```
~ $ aws elb describe-load-balancers --load-balancer-name carrd12
{
    "LoadBalancerDescriptions": [
        {
            "LoadBalancerName": "carrd12",
            "DNSName": "carrd12-1091338731.us-east-1.elb.amazonaws.com",
            "CanonicalHostedZoneName": "carrd12-1091338731.us-east-1.elb.amazonaws.com",
            "CanonicalHostedZoneNameID": "Z35SXDOTRQ7X7K",
            "ListenerDescriptions": [
                {
                    "Listener": {
                        "Protocol": "HTTP",
                        "LoadBalancerPort": 80,
                        "InstanceProtocol": "HTTP",
                        "InstancePort": 80
                    },
                    "PolicyNames": []
                }
            ],
            "Policies": {
                "AppCookieStickinessPolicies": [],
                "LBCookieStickinessPolicies": [],
                "OtherPolicies": []
            },
            "BackendServerDescriptions": [],
            "AvailabilityZones": [
                "us-east-1c"
            ],
:
```

```
            "BackendServerDescriptions": [],
            "AvailabilityZones": [
                "us-east-1c"
            ],
            "Subnets": [
                "subnet-09df9a6645aea6ede"
            ],
            "VPCId": "vpc-0cb3e30db617f6175",
            "Instances": [],
            "HealthCheck": {
                "Target": "TCP:80",
                "Interval": 30,
                "Timeout": 5,
                "UnhealthyThreshold": 2,
                "HealthyThreshold": 10
            },
            "SourceSecurityGroup": {
                "OwnerAlias": "881490118823",
                "GroupName": "carrd12"
            },
            "SecurityGroups": [
                "sg-0b1cfa43fd7ea602b"
            ],
            "CreatedTime": "2025-04-20T23:07:42.010000+00:00",
            "Scheme": "internet-facing"
        }
    ]
}
(END)
```

4.Create a file named apache-install with the following content:

```
#!/bin/bash
# Prevent apt from bringing up any interactive queries
export DEBIAN_FRONTEND=noninteractive
# Linux distribution update, and Apache installation
apt update -y
apt full-upgrade -y
apt install apache2 -y
apt autoremove
/etc/init.d/apache2 restart
```

5.Run the following command to create two instances:

aws ec2 run-instances --image-id ami-09e67e426f25ce0d7 --count 2 --instance-type t3.micro --key-name your_username-key --security-group-ids  your_security_group_id --subnet your_subnet_id --user-data file://apache-install

**What is the first instance ID? What is the second instance ID?**
i-06964a23b7946bc86
i-0bc405562bf0e5fac

```
~ $ aws ec2 run-instances --image-id ami-09e67e426f25ce0d7 --count 2 --instance-type t3.micro --key-name carrd12-key --security-group-ids sg-0b1cfa43fd7ea602b --subnet subnet-09df9a6645aea6ede --user-data file://apache-install
{
    "ReservationId": "r-02188cd0e60618b6d",
    "OwnerId": "881490118823",
    "Groups": [],
    "Instances": [
        {
            "Architecture": "x86_64",
            "BlockDeviceMappings": [],
            "ClientToken": "2ba592c3-fde5-4340-a6e3-97f6b488fa54",
            "EbsOptimized": false,
            "EnaSupport": true,
            "Hypervisor": "xen",
            "NetworkInterfaces": [
                {
                    "Attachment": {
                        "AttachTime": "2025-04-20T23:19:03+00:00",
                        "AttachmentId": "eni-attach-012c69e3b38fea08a",
                        "DeleteOnTermination": true,
                        "DeviceIndex": 0,
                        "Status": "attaching",
                        "NetworkCardIndex": 0
                    },
                    "Description": "",
                    "Groups": [
                        {
                            "GroupId": "sg-0b1cfa43fd7ea602b",
                            "GroupName": "carrd12"
```

```
                    "GroupName": "carrd12"
                }
            ],
            "Ipv6Addresses": [],
            "MacAddress": "12:1e:a7:e8:e1:af",
            "NetworkInterfaceId": "eni-0307d7408c289c20d",
            "OwnerId": "881490118823",
            "PrivateDnsName": "ip-10-6-8-13.ec2.internal",
            "PrivateIpAddress": "10.6.8.13",
            "PrivateIpAddresses": [
                {
                    "Primary": true,
                    "PrivateDnsName": "ip-10-6-8-13.ec2.internal",
                    "PrivateIpAddress": "10.6.8.13"
                }
            ],
            "SourceDestCheck": true,
            "Status": "in-use",
            "SubnetId": "subnet-09df9a6645aea6ede",
            "VpcId": "vpc-0cb3e30db617f6175",
            "InterfaceType": "interface",
            "Operator": {
                "Managed": false
            }
        }
    ],
    "RootDeviceName": "/dev/sda1",
    "RootDeviceType": "ebs",
:
```

```
                "RootDeviceType": "ebs",
                "SecurityGroups": [
                    {
                        "GroupId": "sg-0b1cfa43fd7ea602b",
                        "GroupName": "carrd12"
                    }
                ],
                "SourceDestCheck": true,
                "StateReason": {
                    "Code": "pending",
                    "Message": "pending"
                },
                "VirtualizationType": "hvm",
                "CpuOptions": {
                    "CoreCount": 1,
                    "ThreadsPerCore": 2
                },
                "CapacityReservationSpecification": {
                    "CapacityReservationPreference": "open"
                },
                "MetadataOptions": {
                    "State": "pending",
                    "HttpTokens": "optional",
                    "HttpPutResponseHopLimit": 1,
                    "HttpEndpoint": "enabled",
                    "HttpProtocolIpv6": "disabled",
                    "InstanceMetadataTags": "disabled"
                },
:
```

```
            },
            "EnclaveOptions": {
                "Enabled": false
            },
            "PrivateDnsNameOptions": {
                "HostnameType": "ip-name",
                "EnableResourceNameDnsARecord": false,
                "EnableResourceNameDnsAAAARecord": false
            },
            "MaintenanceOptions": {
                "AutoRecovery": "default"
            },
            "CurrentInstanceBootMode": "legacy-bios",
            "Operator": {
                "Managed": false
            },
            "InstanceId": "i-06964a23b7946bc86",
            "ImageId": "ami-09e67e426f25ce0d7",
            "State": {
                "Code": 0,
                "Name": "pending"
            },
            "PrivateDnsName": "ip-10-6-8-13.ec2.internal",
            "PublicDnsName": "",
            "StateTransitionReason": "",
            "KeyName": "carrd12-key",
            "AmiLaunchIndex": 0,
            "ProductCodes": [],
:
```

```
            "ProductCodes": [],
            "InstanceType": "t3.micro",
            "LaunchTime": "2025-04-20T23:19:03+00:00",
            "Placement": {
                "GroupName": "",
                "Tenancy": "default",
                "AvailabilityZone": "us-east-1c"
            },
            "Monitoring": {
                "State": "disabled"
            },
            "SubnetId": "subnet-09df9a6645aea6ede",
            "VpcId": "vpc-0cb3e30db617f6175",
            "PrivateIpAddress": "10.6.8.13"
        },
        {
            "Architecture": "x86_64",
            "BlockDeviceMappings": [],
            "ClientToken": "2ba592c3-fde5-4340-a6e3-97f6b488fa54",
            "EbsOptimized": false,
            "EnaSupport": true,
            "Hypervisor": "xen",
            "NetworkInterfaces": [
                {
                    "Attachment": {
                        "AttachTime": "2025-04-20T23:19:03+00:00",
                        "AttachmentId": "eni-attach-08a635480125ee143",
                        "DeleteOnTermination": true,
:
```

```
                "DeleteOnTermination": true,
                "DeviceIndex": 0,
                "Status": "attaching",
                "NetworkCardIndex": 0
            },
            "Description": "",
            "Groups": [
                {
                    "GroupId": "sg-0b1cfa43fd7ea602b",
                    "GroupName": "carrd12"
                }
            ],
            "Ipv6Addresses": [],
            "MacAddress": "12:c5:7d:46:99:5b",
            "NetworkInterfaceId": "eni-089d5c9cd2da777d0",
            "OwnerId": "881490118823",
            "PrivateDnsName": "ip-10-6-8-5.ec2.internal",
            "PrivateIpAddress": "10.6.8.5",
            "PrivateIpAddresses": [
                {
                    "Primary": true,
                    "PrivateDnsName": "ip-10-6-8-5.ec2.internal",
                    "PrivateIpAddress": "10.6.8.5"
                }
            ],
            "SourceDestCheck": true,
            "Status": "in-use",
            "SubnetId": "subnet-09df9a6645aea6ede",
:
```

```
                "SubnetId": "subnet-09df9a6645aea6ede",
                "VpcId": "vpc-0cb3e30db617f6175",
                "InterfaceType": "interface",
                "Operator": {
                    "Managed": false
                }
            }
        ],
        "RootDeviceName": "/dev/sda1",
        "RootDeviceType": "ebs",
        "SecurityGroups": [
            {
                "GroupId": "sg-0b1cfa43fd7ea602b",
                "GroupName": "carrd12"
            }
        ],
        "SourceDestCheck": true,
        "StateReason": {
            "Code": "pending",
            "Message": "pending"
        },
        "VirtualizationType": "hvm",
        "CpuOptions": {
            "CoreCount": 1,
            "ThreadsPerCore": 2
        },
        "CapacityReservationSpecification": {
            "CapacityReservationPreference": "open"
:
```

```
            "CapacityReservationPreference": "open"
        },
        "MetadataOptions": {
            "State": "pending",
            "HttpTokens": "optional",
            "HttpPutResponseHopLimit": 1,
            "HttpEndpoint": "enabled",
            "HttpProtocolIpv6": "disabled",
            "InstanceMetadataTags": "disabled"
        },
        "EnclaveOptions": {
            "Enabled": false
        },
        "PrivateDnsNameOptions": {
            "HostnameType": "ip-name",
            "EnableResourceNameDnsARecord": false,
            "EnableResourceNameDnsAAAARecord": false
        },
        "MaintenanceOptions": {
            "AutoRecovery": "default"
        },
        "CurrentInstanceBootMode": "legacy-bios",
        "Operator": {
            "Managed": false
        },
        "InstanceId": "i-0bc405562bf0e5fac",
        "ImageId": "ami-09e67e426f25ce0d7",
        "State": {
:
```

```
            "InstanceId": "i-0bc405562bf0e5fac",
            "ImageId": "ami-09e67e426f25ce0d7",
            "State": {
                "Code": 0,
                "Name": "pending"
            },
            "PrivateDnsName": "ip-10-6-8-5.ec2.internal",
            "PublicDnsName": "",
            "StateTransitionReason": "",
            "KeyName": "carrd12-key",
            "AmiLaunchIndex": 1,
            "ProductCodes": [],
            "InstanceType": "t3.micro",
            "LaunchTime": "2025-04-20T23:19:03+00:00",
            "Placement": {
                "GroupName": "",
                "Tenancy": "default",
                "AvailabilityZone": "us-east-1c"
            },
            "Monitoring": {
                "State": "disabled"
            },
            "SubnetId": "subnet-09df9a6645aea6ede",
            "VpcId": "vpc-0cb3e30db617f6175",
            "PrivateIpAddress": "10.6.8.5"
        }
    ]
}
(END)
```


6.Register two instances with your load balancer with the following command:

aws elb register-instances-with-load-balancer --load-balancer-name your_username --instances first_instance_ID second_instance_ID

**What is your command and its complete output?**

```
~ $ aws elb register-instances-with-load-balancer --load-balancer-name carrd12 --instances i-06964a23b7946bc86 i-0bc405562bf0e5fac
{
    "Instances": [
        {
            "InstanceId": "i-06964a23b7946bc86"
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac"
        }
    ]
}
~ $
```

7.Wait for 20 minutes and then describe the state of the instances with respect to the load balancer with the following command:

aws elb describe-instance-health --load-balancer-name your_username

**What is your command and its complete output?**

```
~ $ aws elb describe-instance-health --load-balancer-name carrd12
{
    "InstanceStates": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
:...skipping...
{
    "InstanceStates": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        }
```

```
:...skipping...
{
    "InstanceStates": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        }
    ]
:...skipping...
{
    "InstanceStates": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "State": "InService",
            "ReasonCode": "N/A",
```

```
                "Description": "N/A"
            }
        ]
:...skipping...
{
    "InstanceStates": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        }
    ]
}
~
(END)...skipping...
{
    "InstanceStates": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
```

```
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        }
    ]
}
~
~
(END)...skipping...
{
    "InstanceStates": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        }
    ]
}
~
```

```
(END)...skipping...
{
    "InstanceStates": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        }
    ]
}
~
~
~
~
(END)...skipping...
(END)...skipping...
{
    "InstanceStates": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
            "ReasonCode": "N/A",
```

```
            "Description": "N/A"
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        }
    ]
}
~
~
~
~
~
(END)...skipping...
{
    "InstanceStates": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
```

```
        }
    ]
}
~
~
~
~
~
~
(END)...skipping...
{
    "InstanceStates": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        }
    ]
}
~
~
~
```

```
(END)...skipping...
{
    "InstanceStates": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        }
    ]
}
~
~
~
~
~
~
~
~
(END)...skipping...
(END)...skipping...
{
    "InstanceStates": [
```

```
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        }
    ]
}
~
~
~
~
~
~
~
~
~
(END)...skipping...
{
    "InstanceStates": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
```


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```
            "ReasonCode": "N/A",
            "Description": "N/A"
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
        }
    ]
}
~
~
~
~
~
~
~
~
~
~
(END)...skipping...
{
    "InstanceStates": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "State": "InService",
            "ReasonCode": "N/A",
            "Description": "N/A"
```

```
                "ReasonCode": "N/A",
                "Description": "N/A"
            },
            {
                "InstanceId": "i-0bc405562bf0e5fac",
                "State": "InService",
                "ReasonCode": "N/A",
                "Description": "N/A"
            }
        ]
}
~
~
~
~
~
~
~
~
~
~
~
(END)
```

Note: If the state of your EC2 instances is OutOfService, please wait for several minutes and try to run the above command again. You should see the state changes to InService.

8.Create a bucket to store ELB access logs via the following command:

aws s3api create-bucket --region us-east-1 --bucket your_username-elb-log

**What is your command and its complete output?**

```
~ $ aws s3api create-bucket --region us-east-1 --bucket carrd12-elb-log
{
    "Location": "/carrd12-elb-log"
}
~ $
```

9.You may still have a file named policy.json from a previous lab. Because we need to recreate this file, delete the old one with this command: rm policy.json.  Create a json file named policy.json with the following content:

{

```
 "Version": "2012-10-17",
 "Statement": [
  {
   "Effect": "Allow",
   "Principal": {
    "AWS": "arn:aws:iam::127311923021:root"
   },
   "Action": "s3:PutObject",
   "Resource": "arn:aws:s3:::your_username-elb-log/*"
  },
  {
   "Effect": "Allow",
   "Principal": {
    "Service": "delivery.logs.amazonaws.com"
   },
   "Action": "s3:PutObject",
   "Resource": "arn:aws:s3:::your_username-elb-log/*",
   "Condition": {
    "StringEquals": {
     "s3:x-amz-acl": "bucket-owner-full-control"
    }
   }
  },
  {
   "Effect": "Allow",
   "Principal": {
    "Service": "delivery.logs.amazonaws.com"
   },
   "Action": "s3:GetBucketAcl",
   "Resource": "arn:aws:s3:::your_username-elb-log"
  }
 ]
}
```

10.Put the policy for the bucket with the following command:

aws s3api put-bucket-policy --bucket your_username-elb-log --policy file://policy.json

**What is your command?**

```
~ $ aws s3api put-bucket-policy --bucket carrd12-elb-log --policy file://policy.json
~ $
```

11.Create a json file named enable.json with the following content:

```
{
 "AccessLog": {
  "Enabled": true,
  "S3BucketName": "your_username-elb-log",
  "EmitInterval": 5,
  "S3BucketPrefix": "elb"
 }
}
```

**How often does the ELB publish its access log?**
**Every 5 minutes**

12.Enable ELB's access log via the following command:

aws elb modify-load-balancer-attributes --load-balancer-name your_username --load-balancer-attributes file://enable.json

**What is your command and its complete output?**

```
~ $ aws elb modify-load-balancer-attributes --load-balancer-name carrd12 --load-balancer-attributes file://enable.json
{
    "LoadBalancerName": "carrd12",
    "LoadBalancerAttributes": {
        "AccessLog": {
            "Enabled": true,
            "S3BucketName": "carrd12-elb-log",
            "EmitInterval": 5,
            "S3BucketPrefix": "elb"
        }
    }
}
~ $
```

13.Use the ab command to conduct a test via the following command:

ab -c 50 -n 1000 http://your_load_balancer_domain_name/

**What is the complete output?**

```
~ $ ab -c 50 -n 1000 http://carrd12-1091338731.us-east-1.elb.amazonaws.com/
This is ApacheBench, Version 2.3 <$Revision: 1913912 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking carrd12-1091338731.us-east-1.elb.amazonaws.com (be patient)
Completed 100 requests
Completed 200 requests
Completed 300 requests
Completed 400 requests
Completed 500 requests
Completed 600 requests
Completed 700 requests
Completed 800 requests
Completed 900 requests
Completed 1000 requests
Finished 1000 requests


Server Software:        Apache/2.4.41
Server Hostname:        carrd12-1091338731.us-east-1.elb.amazonaws.com
Server Port:            80

Document Path:          /
Document Length:        10918 bytes

Concurrency Level:      50
Time taken for tests:   0.190 seconds
Complete requests:      1000
```

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```
Failed requests:        0
Total transferred:      11192000 bytes
HTML transferred:       10918000 bytes
Requests per second:    5253.70 [#/sec] (mean)
Time per request:       9.517 [ms] (mean)
Time per request:       0.190 [ms] (mean, across all concurrent requests)
Transfer rate:          57421.31 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        1    1   0.3      1       3
Processing:     3    8   3.0      7      22
Waiting:        2    8   2.9      7      22
Total:          4    9   3.0      8      23

Percentage of the requests served within a certain time (ms)
  50%      8
  66%      9
  75%     10
  80%     11
  90%     12
  95%     17
  98%     19
  99%     19
 100%     23 (longest request)
~ $
```

This is the metric measured on the client side.

If you have not installed the ab tool on your CloudShell, run these commands:

sudo su
yum install httpd-tools -y
exit

14.Now let's look at metrics measured on the server side.  ELB access logs capture detailed information about each request sent to your load balancer. Each log contains information such as the time the request was received, the client's IP address, latencies, request paths, and server responses. You can use these access logs to analyze traffic patterns and troubleshoot issues. Let's check your ELB access log. Wait for 10 minutes and then run the following command to download your ELB's access log:

aws s3 sync s3://your_username-elb-log/ ./

**What is your output?** The command should create log files under a hierarchy of directories (nested directories). Here is an example of my downloaded log file inside nested directories under the current directory.

```
~ $ aws s3 sync s3://carrd12-elb-log/ ./
download: s3://carrd12-elb-log/elb/AWSLogs/881490118823/ELBAccessLogTestFile to elb/AWSLogs/881490118823/ELBAccessLogTestFile
download: s3://carrd12-elb-log/elb/AWSLogs/881490118823/elasticloadbalancing/us-east-1/2025/04/21/881490118823_elasticloadbalancing_u
s-east-1_carrd12_20250421T0025Z__wakyi7dl.log to elb/AWSLogs/881490118823/elasticloadbalancing/us-east-1/2025/04/21/881490118823_elas
ticloadbalancing_us-east-1_carrd12_20250421T0025Z__wakyi7dl.log
~ $
```

./elb/AWSLogs/165786971191/elasticloadbalancing/us-east-1/2022/06/02/165786971191_elasticloadbalancing_us-east-1_haow1_20220602T1155Z__4noet3r0.log

Please read the Access log files section at
https://docs.aws.amazon.com/elasticloadbalancing/latest/classic/access-log-collection.html

15.The s3 sync command may download multiple access log files. Please use the access log file containing your_load_balancer_name.

**What are the first 10 log entries in your access log file?** (You can use vi to open your log file or use the head command, head -10 your_log_filename)
s3://carrd12-elb-log/elb/AWSLogs/881490118823/elasticloadbalancing/us-east-1/2025/04/21/881490118823_elasticloadbalancing_us-east-1_carrd12_20250421T0025Z__wakyi7dl.log

```
2025-04-21T00:25:44.096462Z carrd12 3.80.167.13:34846 10.6.8.5:80 0.000016 0.000687 0.000011 200 200 0 10918 "GET http://carrd12-1091
2025-04-21T00:25:43.914062Z carrd12 3.80.167.13:54156 10.6.8.13:80 0.000027 0.000677 0.000024 200 200 0 10918 "GET http://carrd12-109
1338731.us-east-1.elb.amazonaws.com:80/ HTTP/1.0" "ApacheBench/2.3" - -
2025-04-21T00:25:43.918526Z carrd12 3.80.167.13:54160 10.6.8.5:80 0.000023 0.000782 0.000012 200 200 0 10918 "GET http://carrd12-1091
338731.us-east-1.elb.amazonaws.com:80/ HTTP/1.0" "ApacheBench/2.3" - -
2025-04-21T00:25:43.918672Z carrd12 3.80.167.13:54176 10.6.8.13:80 0.000016 0.000809 0.000014 200 200 0 10918 "GET http://carrd12-109
1338731.us-east-1.elb.amazonaws.com:80/ HTTP/1.0" "ApacheBench/2.3" - -
2025-04-21T00:25:43.918789Z carrd12 3.80.167.13:54190 10.6.8.13:80 0.000016 0.000472 0.000017 200 200 0 10918 "GET http://carrd12-109
1338731.us-east-1.elb.amazonaws.com:80/ HTTP/1.0" "ApacheBench/2.3" - -
2025-04-21T00:25:43.919091Z carrd12 3.80.167.13:54224 10.6.8.5:80 0.000014 0.00055 0.000014 200 200 0 10918 "GET http://carrd12-10913
38731.us-east-1.elb.amazonaws.com:80/ HTTP/1.0" "ApacheBench/2.3" - -
2025-04-21T00:25:43.918940Z carrd12 3.80.167.13:54194 10.6.8.13:80 0.000017 0.000808 0.000013 200 200 0 10918 "GET http://carrd12-109
1338731.us-east-1.elb.amazonaws.com:80/ HTTP/1.0" "ApacheBench/2.3" - -
2025-04-21T00:25:43.919212Z carrd12 3.80.167.13:54220 10.6.8.13:80 0.000032 0.000989 0.000018 200 200 0 10918 "GET http://carrd12-109
1338731.us-east-1.elb.amazonaws.com:80/ HTTP/1.0" "ApacheBench/2.3" - -
2025-04-21T00:25:43.919383Z carrd12 3.80.167.13:54228 10.6.8.5:80 0.000014 0.002614 0.000092 200 200 0 10918 "GET http://carrd12-1091
338731.us-east-1.elb.amazonaws.com:80/ HTTP/1.0" "ApacheBench/2.3" - -
2025-04-21T00:25:43.919732Z carrd12 3.80.167.13:54238 10.6.8.5:80 0.000022 0.002629 0.000013 200 200 0 10918 "GET http://carrd12-1091
338731.us-east-1.elb.amazonaws.com:80/ HTTP/1.0" "ApacheBench/2.3" - -
2025-04-21T00:25:43.919875Z carrd12 3.80.167.13:54244 10.6.8.13:80 0.000268 0.002308 0.00001 200 200 0 10918 "GET http://carrd12-1091
338731.us-east-1.elb.amazonaws.com:80/ HTTP/1.0" "ApacheBench/2.3" - -
2025-04-21T00:25:43.920254Z carrd12 3.80.167.13:54254 10.6.8.5:80 0.000018 0.002493 0.000044 200 200 0 10918 "GET http://carrd12-1091
338731.us-east-1.elb.amazonaws.com:80/ HTTP/1.0" "ApacheBench/2.3" - -
2025-04-21T00:25:43.921125Z carrd12 3.80.167.13:54210 10.6.8.13:80 0.00002 0.001756 0.000011 200 200 0 10918 "GET http://carrd12-1091
338731.us-east-1.elb.amazonaws.com:80/ HTTP/1.0" "ApacheBench/2.3" - -
2025-04-21T00:25:43.921218Z carrd12 3.80.167.13:54262 10.6.8.5:80 0.000011 0.002781 0.000009 200 200 0 10918 "GET http://carrd12-1091
338731.us-east-1.elb.amazonaws.com:80/ HTTP/1.0" "ApacheBench/2.3" - -
@@@
                                                                                                   1,1           Top
```

Please look at the Access log entries section at
https://docs.aws.amazon.com/elasticloadbalancing/latest/classic/access-log-collection.html and explain one log entry.

```
2025-04-21T00:25:44.097059Z carrd12 3.80.167.13:34902 10.6.8.13:80 0.000012 0.005248 0.000017 200 200 0 10918 "GET http://carrd12-109
1338731.us-east-1.elb.amazonaws.com:80/ HTTP/1.0" "ApacheBench/2.3" - -
```

This ELB log entry shows a successful GET request made by the ApacheBench tool to the `carrd12` load balancer. The request came from an external IP and was routed to a backend EC2 instance, which responded with a 200 OK status. The entire transaction was completed very quickly, with minimal processing time, and the response sent to the client was about 10KB in size.

**Part 2: Cleanup**

16.Please remove your EC2 instances. **What commands did you run?**

```
~ $ aws ec2 terminate-instances  --instance-ids i-06964a23b7946bc86 i-0bc405562bf0e5fac
{
    "TerminatingInstances": [
        {
            "InstanceId": "i-06964a23b7946bc86",
            "CurrentState": {
                "Code": 32,
                "Name": "shutting-down"
            },
            "PreviousState": {
                "Code": 16,
                "Name": "running"
            }
        },
        {
            "InstanceId": "i-0bc405562bf0e5fac",
            "CurrentState": {
                "Code": 32,
                "Name": "shutting-down"
            },
            "PreviousState": {
                "Code": 16,
                "Name": "running"
            }
        }
    ]
}
~ $
```

17.Please remove your s3 bucket. **What command did you run?**

```
~ $ aws s3 rb s3://carrd12-elb-log/ --force
delete: s3://carrd12-elb-log/elb/AWSLogs/881490118823/ELBAccessLogTestFile
delete: s3://carrd12-elb-log/elb/AWSLogs/881490118823/elasticloadbalancing/us-east-1/2025/04/21/881490118823_elasticloadbalancing_us-east-1_carrd12_20250421T0040Z__28vvp0h8.log
delete: s3://carrd12-elb-log/elb/AWSLogs/881490118823/elasticloadbalancing/us-east-1/2025/04/21/881490118823_elasticloadbalancing_us-east-1_carrd12_20250421T0025Z__wakyi7dl.log
remove_bucket: carrd12-elb-log
~ $
```

18.Please remove your ELB. **What command did you run?**

```
~ $ aws elb delete-load-balancer --load-balancer-name carrd12
~ $
```

Submission instructions
After completing this work, submit this word document with your answers to Canvas. In your word document, your answers should be in **bold** print. At the end of your word document, you should write a few sentences that explain the goals of this lab and a paragraph or two that summarizes (high level) the steps one needs to take in order to complete those goals.

My lab's objective was to enable and install Elastic Load Balancer (ELB) access logging using AWS CLI commands. This would enable me to record comprehensive request data for monitoring and analysis. It helps in security auditing, performance optimization, and troubleshooting. You used the aws elb modify-load-balancer-attributes command to enable logging and specify the log delivery bucket and interval, created or identified an S3 bucket for log storage, and updated the relevant bucket policy to permit ELB to write logs in order to finish the experiment.